SAP MM PROJECT REPORT

Procure-to-Pay (P2P) — Full Purchasing Cycle

Blueprint & Step-by-Step Process Document

Submitted To:
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1. Problem Statement

In many organisations, procurement is handled manually through disconnected systems — causing delays, errors, poor transparency, and weak coordination. The result: incorrect inventory, delayed payments, and financial discrepancies.
An integrated system is needed to automate the P2P cycle and ensure accuracy, real-time tracking, and seamless coordination between procurement and finance.

2. Solution & Technology Stack

SAP MM automates the entire P2P cycle — PR to vendor payment — in a single integrated system.

- Three-way match (PO ↔ GR ↔ Invoice) prevents erroneous or duplicate payments.
- Real-time FI integration auto-posts accounting documents at every step.
- Automated F110 runs ensure on-time, auditable vendor payments per agreed terms.

Component	Technology / Tool
ERP Platform	SAP ECC 6.0 / S/4HANA
Core Module	SAP MM + SAP FI (Financial Accounting)
Reporting & Output	ALV Reports (ME2M, MB51) + SAP SmartForms (PO Print)

3. Project Blueprint

3.1 Organisation Structure — ABC Manufacturing Pvt. Ltd.

Org Unit	Code / ID	Description
Company Code	1000	ABC Manufacturing — Head Office, Bhubaneswar
Plant / Storage Location	1100 / SL01	Main Production Plant / Raw Material Store
Purchase Org / Group	PO01 / PG1	Central Purchasing / Direct Materials Buyers

3.2 End-to-End Process Flow

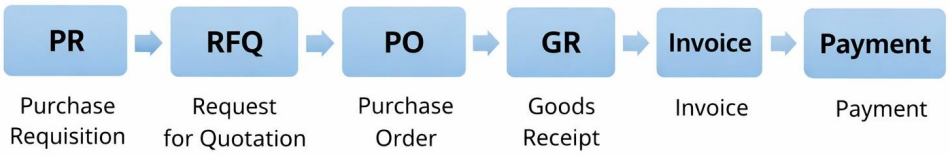

Figure 1: Procure-to-Pay (P2P) Process Flow — PR → RFQ → PO → GR → Invoice → Payment

3.3 Key Transaction Codes — P2P Scope

Step	T-Code	Activity
1. Purchase Requisition	ME51N	Internal material demand trigger
2. Request for Quotation	ME41 / ME47	Create RFQ, enter vendor quotes
3. Price Comparison	ME49	Compare & select best vendor
4. Purchase Order	ME21N	Legally binding PO to vendor
5. Goods Receipt	MIGO	Post receipt, Movement Type 101
6. Invoice Verification	MIRO	3-way match: PO – GR – Invoice
7. Payment Run	F110	Auto payment, clear vendor

4. Step-by-Step Process

Step 1 — Purchase Requisition (ME51N)

- ME51N → Doc Type NB → Account Assignment K (cost centre) or blank (stock)
- Enter Material, Qty, Delivery Date, Plant 1100, SLoc SL01, Purchase Group PG1 → Save → PR No. generated

Step 2 — Request for Quotation (ME41 / ME49)

- ME41: Create RFQ → Purch. Org PO01 → add 3+ vendors → ME47: enter vendor quoted prices
- ME49: Price Comparison List → select lowest bidder → ME48: reject unsuccessful vendors

Step 3 — Create Purchase Order (ME21N)

- ME21N: PO Type NB → Header: Vendor, PO01, PG1, Co. Code 1000
- Item: Material, Qty, Plant 1100, Net Price, Tax Code V0 → Delivery Tab: tolerance 5%, GR-Based IV = active
- Conditions Tab: Verify PBXX Gross Price & Z101 Standard Price → Save → PO No. generated → send to vendor

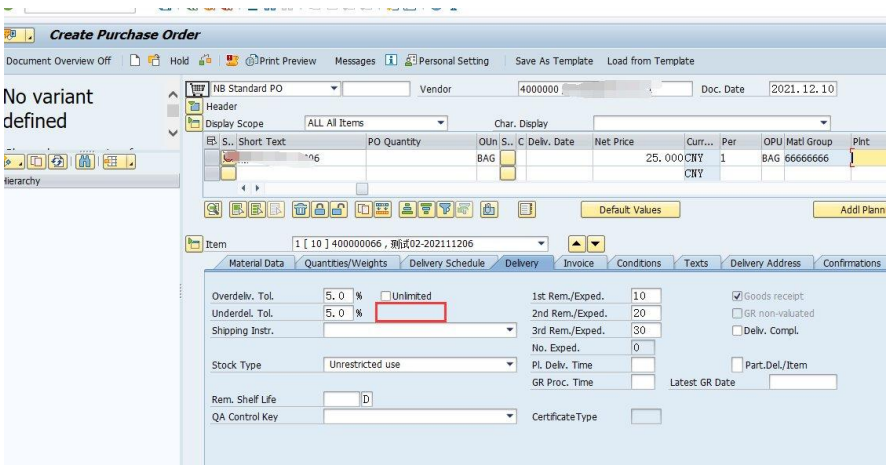

Figure 2: ME21N — Create PO, Delivery Tab & Pricing Conditions

Step 4 — Goods Receipt (MIGO)

- MIGO → Goods Receipt → Purchase Order → Enter PO No. → verify material, qty, SLoc SL01
- Movement Type 101 auto-assigned → Check Item OK → Post → Material Document + FI Document created

Accounting: Dr Inventory A/c | Cr GR/IR Clearing A/c

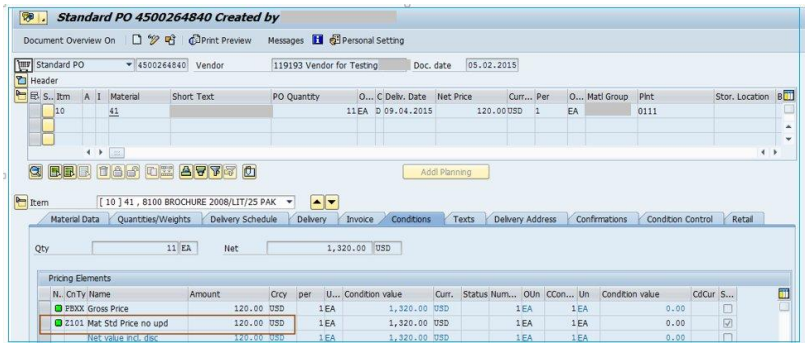

Figure 3: MIGO — Goods Receipt against Purchase Order

Step 5 — Invoice Verification (MIRO)

- MIRO → Transaction: 1 Invoice → Invoice Date, Posting Date, Co. Code → PO Reference Tab: PO No.
- System auto-proposes qty & amounts → verify Balance = 0.00 (3-way match: PO ↔ GR ↔ Invoice)
- Within tolerance → auto-post | Exceeds tolerance → invoice blocked; release via MRBR

Accounting: Dr GR/IR Clearing A/c | Cr Vendor A/c

Figure 4: MIRO — Enter Incoming Invoice, Company Code 0001

Step 6 — Automatic Payment Run (F110)

- F110: Run Date + ID Key → Parameters: Co. Code 1000, Pay Method, Vendor range
- Additional Log: enable Due Date check & Line items → Execute Proposal → Execute Payment Run

Accounting: Dr Vendor A/c | Cr Bank A/c

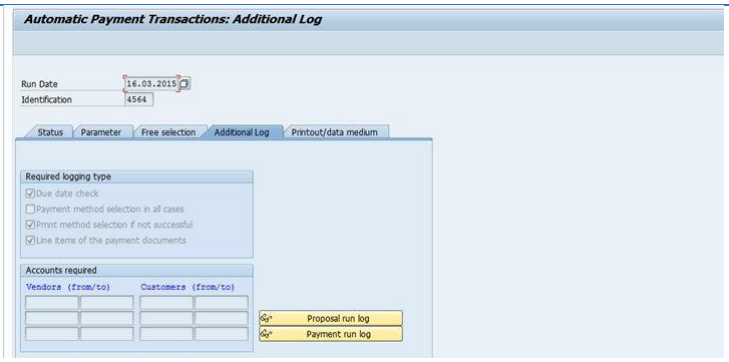

Figure 5: F110 — Automatic Payment Transactions, Additional Log

5. FI Integration & Document Flow

Event	T-Code	Accounting Entry
Goods Receipt	MIGO	Dr Inventory A/c / Cr GR-IR Clearing A/c
Invoice Verification	MIRO	Dr GR-IR Clearing / Cr Vendor A/c
Payment Run	F110	Dr Vendor A/c / Cr Bank A/c

Document	T-Code	Links To
Purchase Requisition	ME51N	Converted to PO via ME21N
Purchase Order	ME21N	References PR & accepted RFQ
Material Document (GR)	MIGO	References PO; updates stock
Accounting Document	MIRO	References PO + GR (3-way match)
Payment Document	F110	Clears open vendor item from MIRO

6. Unique Points & Future Improvements

Unique Points:

- GR-Based IV: Only allows MIRO posting after physical goods confirmation — prevents early/fraudulent payments.
- Auto Invoice Blocking: Invoices deviating beyond tolerance keys (±5%) are blocked automatically in MIRO.
- Dual Posting at GR: SAP simultaneously creates a Material Document and FI Document — no manual reconciliation.

Future Improvements:

- SRM Integration: Vendor self-service portal for automated RFQ responses and PO acknowledgements.
- SAP Analytics Cloud: Real-time dashboards for spend analysis, vendor scorecards and PO cycle KPIs.
- Intelligent RPA: AI invoice scanning to auto-populate MIRO fields and reduce data entry errors.

Conclusion: This project delivers a complete SAP MM P2P cycle for ABC Manufacturing Pvt. Ltd., resolving manual procurement through automation, three-way match control, and real-time FI integration — providing a transparent, scalable, and audit-ready procurement framework.